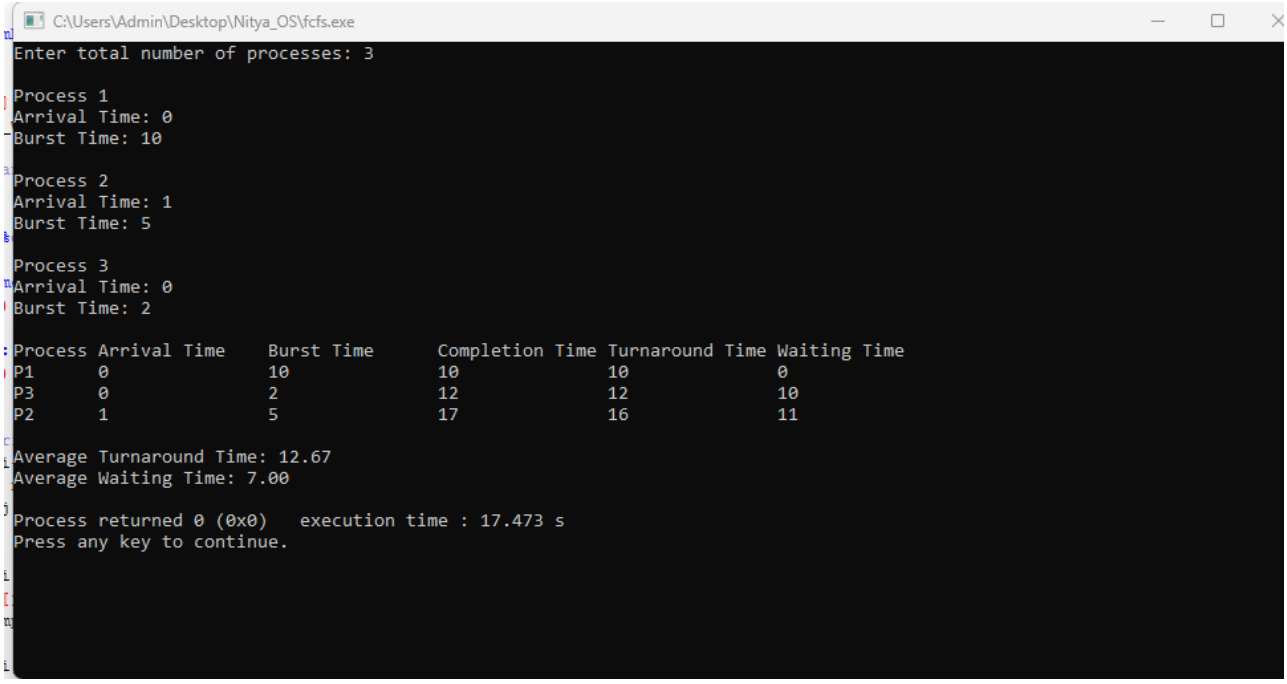
FCFS:

```
//first come first serve

#include <stdio.h>

int main() {
   int n, i, j;

   printf("Enter total number of processes: ");
   scanf("%d", &n);

   int at[n], bt[n], ct[n], tat[n], wt[n], pid[n];
   float avg_tat = 0, avg_wt = 0;

   // Input Arrival Time and Burst Time
   for(i = 0; i < n; i++) {
      pid[i] = i + 1;
      printf("\nProcess %d\n", pid[i]);

      printf("Arrival Time: ");
      scanf("%d", &at[i]);

      printf("Burst Time: ");
      scanf("%d", &bt[i]);
   }

   // Sort processes by Arrival Time
   for(i = 0; i < n - 1; i++) {
      for(j = i + 1; j < n; j++) {
         if(at[i] > at[j]) {
```

```
            int temp;

            temp = at[i];
            at[i] = at[j];
            at[j] = temp;

            temp = bt[i];
            bt[i] = bt[j];
            bt[j] = temp;

            temp = pid[i];
            pid[i] = pid[j];
            pid[j] = temp;
        }
    }
}

int current_time = 0;

// Calculate Completion Time, Turnaround Time, Waiting Time
for(i = 0; i < n; i++) {

    if(current_time < at[i])
        current_time = at[i];

    ct[i] = current_time + bt[i];
    tat[i] = ct[i] - at[i];
    wt[i] = tat[i] - bt[i];

    current_time = ct[i];
```

```
        avg_tat += tat[i];
        avg_wt += wt[i];
    }

    avg_tat /= n;
    avg_wt /= n;

    // Output
    printf("\nProcess\tArrival Time\tBurst Time\tCompletion Time\tTurnaround Time\tWaiting Time\n");

    for(i = 0; i < n; i++) {
        printf("P%d\t%d\t\t%d\t\t%d\t\t%d\t\t%d\n",
            pid[i], at[i], bt[i], ct[i], tat[i], wt[i]);
    }

    printf("\nAverage Turnaround Time: %.2f", avg_tat);
    printf("\nAverage Waiting Time: %.2f\n", avg_wt);

    return 0;
}
```

OUTPUT:


C:\Users\Admin\Desktop\Nitya_OS\fcfs.exe
Enter total number of processes: 3
Process 1
Arrival Time: 0
Burst Time: 10
Process 2
Arrival Time: 1
Burst Time: 5
Process 3
Arrival Time: 0
Burst Time: 2
Process Arrival Time Burst Time Completion Time Turnaround Time Waiting Time
P1 0 10 10 10 0
P3 0 2 12 12 10
P2 1 5 17 16 11
Average Turnaround Time: 12.67
Average Waiting Time: 7.00
Process returned 0 (0x0) execution time : 17.473 s
Press any key to continue.

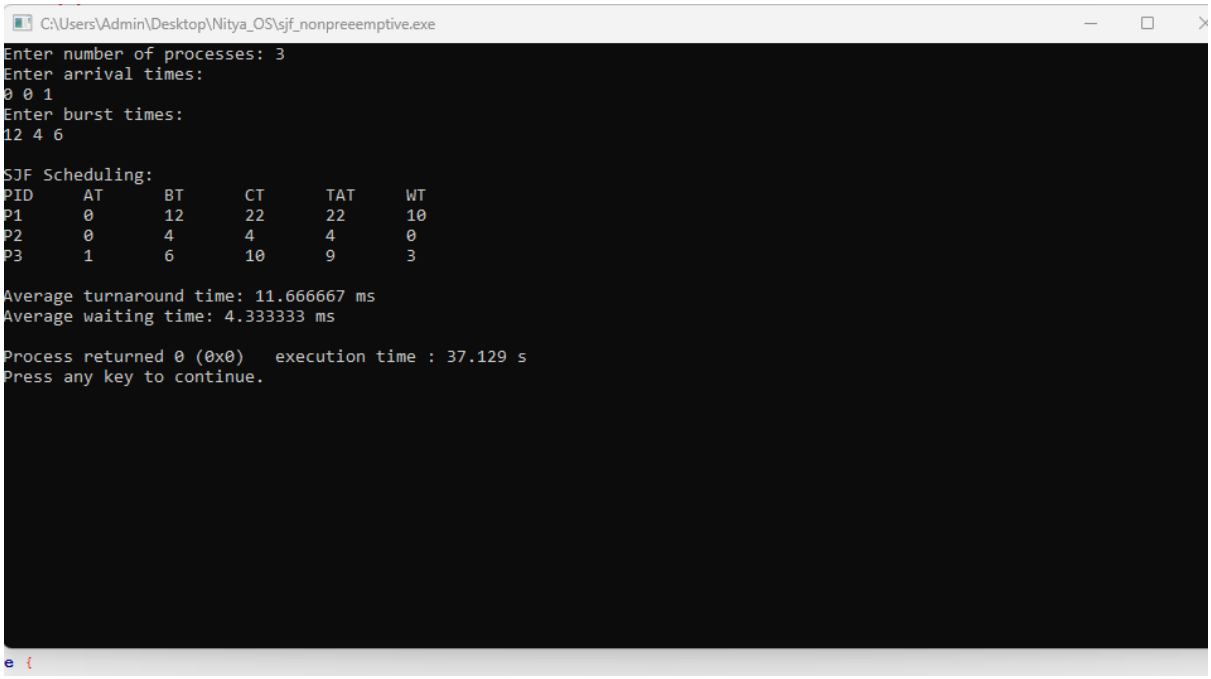
SJF -Non Preemptive:

```
#include <stdio.h>
int main() {
   int n, i, j;

   printf("Enter number of processes: ");
   scanf("%d", &n);

   int at[n], bt[n], ct[n], tat[n], wt[n], pid[n];
   int finished[n];

   float avg_tat = 0, avg_wt = 0;

   // Input arrival times
   printf("Enter arrival times:\n");
   for(i = 0; i < n; i++) {
      scanf("%d", &at[i]);
      pid[i] = i + 1;
      finished[i] = 0;
   }

   // Input burst times
   printf("Enter burst times:\n");
   for(i = 0; i < n; i++) {
      scanf("%d", &bt[i]);
   }

   int completed = 0;
```

```
int current_time = 0;

while(completed < n) {

   int min_bt = 9999;
   int index = -1;

   // Find process with minimum burst time in ready queue
   for(i = 0; i < n; i++) {
      if(at[i] <= current_time && finished[i] == 0) {
         if(bt[i] < min_bt) {
            min_bt = bt[i];
            index = i;
         }
      }
   }

   // If no process is ready
   if(index == -1) {
      current_time++;
   }
   else {
      ct[index] = current_time + bt[index];
      tat[index] = ct[index] - at[index];
      wt[index] = tat[index] - bt[index];

      current_time = ct[index];
      finished[index] = 1;
      completed++;

      avg_tat += tat[index];
```

```
            avg_wt += wt[index];
        }
    }

    avg_tat /= n;
    avg_wt /= n;

    // Output
    printf("\nSJF Scheduling:\n");
    printf("PID\tAT\tBT\tCT\tTAT\tWT\n");

    for(i = 0; i < n; i++) {
        printf("P%d\t%d\t%d\t%d\t%d\t%d\n",
               pid[i], at[i], bt[i], ct[i], tat[i], wt[i]);
    }

    printf("\nAverage turnaround time: %f ms\n", avg_tat);
    printf("Average waiting time: %f ms\n", avg_wt);

    return 0;
}
```

OUTPUT:


C:\Users\Admin\Desktop\Nitya_OS\sjf_nonpreeemptive.exe
Enter number of processes: 3
Enter arrival times:
0 0 1
Enter burst times:
12 4 6
SJF Scheduling:
PID AT BT CT TAT WT
P1 0 12 22 22 10
P2 0 4 4 4 0
P3 1 6 10 9 3
Average turnaround time: 11.666667 ms
Average waiting time: 4.333333 ms
Process returned 0 (0x0) execution time : 37.129 s
Press any key to continue.

```
e {
```

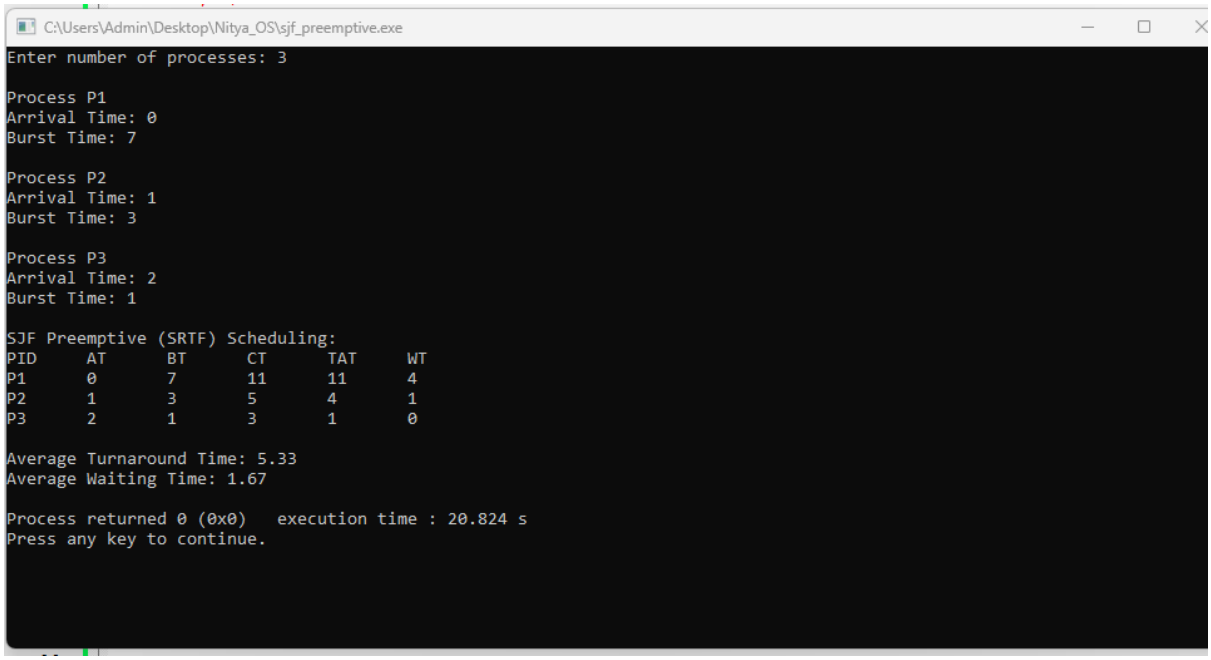
SJF – Preemptive:

```
#include <stdio.h>

int main() {
   int n, i;

   printf("Enter number of processes: ");
   scanf("%d", &n);

   int at[n], bt[n], rt[n], ct[n], tat[n], wt[n], pid[n];

   for(i = 0; i < n; i++) {
      pid[i] = i + 1;
      printf("\nProcess P%d\n", pid[i]);

      printf("Arrival Time: ");
      scanf("%d", &at[i]);

      printf("Burst Time: ");
      scanf("%d", &bt[i]);

      rt[i] = bt[i];   // Remaining time initially equals burst time
   }

   int completed = 0;
   int current_time = 0;
   int min_index;
   int min_rt;
```

```
float avg_tat = 0, avg_wt = 0;

while(completed < n) {

   min_rt = 9999;
   min_index = -1;

   // Find process with smallest remaining time
   for(i = 0; i < n; i++) {
      if(at[i] <= current_time && rt[i] > 0) {
         if(rt[i] < min_rt) {
            min_rt = rt[i];
            min_index = i;
         }
      }
   }

   // If no process is ready
   if(min_index == -1) {
      current_time++;
   }
   else {
      // Execute process for 1 unit
      rt[min_index]--;
      current_time++;

      // If process finishes
      if(rt[min_index] == 0) {
         completed++;

         ct[min_index] = current_time;
```

```
            tat[min_index] = ct[min_index] - at[min_index];
            wt[min_index] = tat[min_index] - bt[min_index];

            avg_tat += tat[min_index];
            avg_wt += wt[min_index];
        }
    }
}

avg_tat /= n;
avg_wt /= n;

// Output
printf("\nSJF Preemptive (SRTF) Scheduling:\n");
printf("PID\tAT\tBT\tCT\tTAT\tWT\n");

for(i = 0; i < n; i++) {
    printf("P%d\t%d\t%d\t%d\t%d\t%d\n",
           pid[i], at[i], bt[i], ct[i], tat[i], wt[i]);
}

printf("\nAverage Turnaround Time: %.2f\n", avg_tat);
printf("Average Waiting Time: %.2f\n", avg_wt);

return 0;
}
```

OUTPUT:


C:\Users\Admin\Desktop\Nitya_OS\sjf_preemptive.exe
Enter number of processes: 3
Process P1
Arrival Time: 0
Burst Time: 7
Process P2
Arrival Time: 1
Burst Time: 3
Process P3
Arrival Time: 2
Burst Time: 1
SJF Preemptive (SRTF) Scheduling:
PID AT BT CT TAT WT
P1 0 7 11 11 4
P2 1 3 5 4 1
P3 2 1 3 1 0
Average Turnaround Time: 5.33
Average Waiting Time: 1.67
Process returned 0 (0x0) execution time : 20.824 s
Press any key to continue.